

비행절차설계 결과보고서

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○ ○ 기 관
○ ○ 부 서

절차명(Title of Procedure) :

절차설계 시작일(Task commences date) :

절차설계담당자(Assign Procedure Designer) :

초안 작성일(Initial Design Ready Date):

수정 횟수 및 수정일자(Revised version number and date):

최종안 작성 및 승인일(Final version ready and approval date):

요약(Executive Summary) :

첨부 : 표준화 문서 보고 서식(예시)

1. ILS APPROACH PROCEDURE(CAT I)
2. ILS APPROACH PROCEDURE(CAT II & III)
3. LOC APPROACH PROCEDURE
4. VOR APPROACH PROCEDURE
5. LNAV APPROACH PROCEDURE
6. LNAV/VNAV APPROACH PROCEDURE
7. CONV SID(DEP)
8. RNAV SID(DEP)
9. CONV STAR
10. RNAV STAR
11. CIRCLING APPROACH COMPUTATION
12. CONV ATS ROUTE
13. RNAV ATS ROUTE

1. ILS APPROACH PROCEDURE(CAT I)

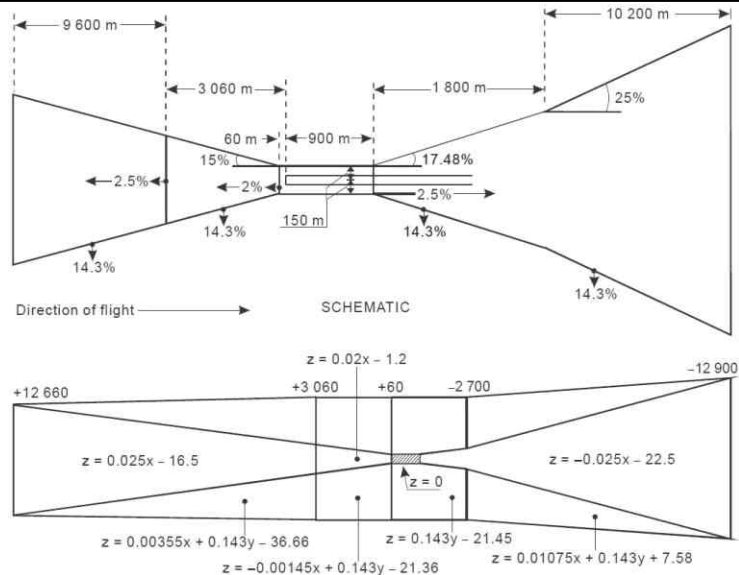
 국토교통부 Ministry of Land, Infrastructure and Transport		ILS APPROACH PROCEDURE					
1. TITLE OF PROCEDURE		2. RWY NUMBER			3. REVISION DATE		
ILS Z RWY 07 CAT I (2.5%)							
4. OPERATING AGENCY		5. NEW OR REVISED			6. APPROVAL DATE		
7. AD INFORMATION							
AD NAME	ICAO IDENTIFIER	AD ELEV (ft)		ARP LAT/LONG		ARP MAG VAR	
8. USE RWY INFORMATION							
RWY NO	DIMENSION (m)	LAT/LONG		THR ELEV (ft)		BEARING(T°/M°)	
9. VISUAL AIDS							
APPROACH LIGHTS		VASIS/PAPI		RUNWAY LIGHTS		OTHER LIGHTS	
NOTE							
10. USE NAVAID INFORMATION							
ILS RDH (ft)			GP ANGLE (°)				
NAME	LAT/LONG		ELEV (ft)		MAG VAR		
11. SPECIFICATION & USE SENSOR							
SPECIFICATION	GNSS		DME/DME		VOR/DME		
12. MAP REFERENCE							
PUBLISHING OFFICE	DATE OF PUBLICATION		MAP NAME		SCALE		
13. PROCEDURE & REVISION SUMMARY							
14. PLAN VIEW OPERATIONAL NOTES							
15. MISSED APPROACH INSTRUCTION							
16. LANDING MINIMA							
CATEGORY		DA (DH)	A	B	C	D	DL
CAT I	FULL						
	ALS INOP						
CAT II							
NOTE							

17. FIX DATA									
TYPE		IDENT		WP/FIX		LAT/LONG			
IAF									
IF									
FAP									
THR									
MATF									
MAHF									
18. TRACK & SEGMENT DATA									
TYPE		IDENT	TRK °M (°T)	DIST (NM)	ALT (ft)	DG (%)	SPEED (kt)		
IAF									
IF									
FAP									
THR									
MATF									
MAHF									
NOTE									
19. MINIMUM STABILIZATION DISTANCE(MSD) - RNAV SEGMENTS ONLY									
SEGMENT		DESIGN DIST(NM)		MSD REQUIRED(NM)		REMAIN(NM)		CONFIRMATION	
20. TURN PARAMETER									
TYPE		IDENT	TURN ANLGE (°)	RADIUS OF TURN[r] (NM)	WIND EFFECT [E90°] (NM)	EARLIEST TURN POINT (NM)		LATEST TURN POINT (NM)	
21. CONTROLLING OBSTACLES									
SEGMENT		ID	NAME	CODE	TYPE	ALTITUDE (ft)		LAT/LONG	
Initial									
Intermediate									
Final1)									
Missed APP (Straight)									
Missed APP (Turning)									
VSS									
NOTE									
22. HOLDING INFORMATION									
IDENT	DIRECTION	ID	NAME	LAT/LONG		ELEV (ft)	MOC (ft)	MIN/MAX ALTITUDE (ft)	SPEED (kt)
23. MINIMUM SECTOR ALTITUDE (MSA)									
NAVAID/WPT				TYPE				IDENT	

TO	FROM	ARC (NM)	ID	NAME	ALT (ft)	LAT/LONG	MOC (ft)	MSA (ft)
NOTE								
24. EXEMPTION CRITERIA								
REQUIRED		REASON						
NOTE								
25. FLIGHT CHECK INFORMATION								
DATE OF FLIGHT CHECK		REASON					REMARKS	
26. CONCLUSION								
27. DEVELOPED BY								
NAME					DATE			
28. APPROVED								
NAME					DATE			
29. REMARKS								

AERONAUTICAL DATA TABULATION	
ILS/LOC Approach to RWY 07 from YUMIN(IAF)	
Fix/point	Coordinates

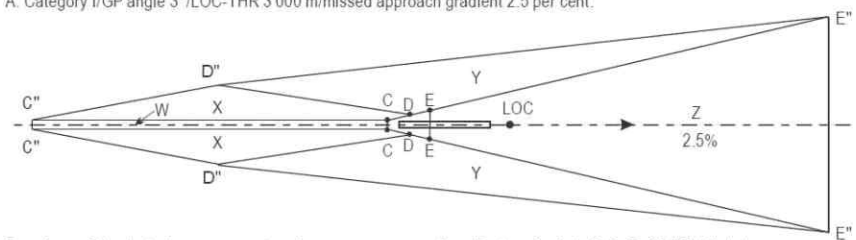
PRECISION COMPUTATIONS for CAT I (2.5%)				
1. COMPUTATION DATA				
THR ELEV(m)	ILS RDH(m)	LOC THR DIST(m)	GP ANGLE(°)	COURSE WIDTH at THR(m)
MISSED APP CG(%)	AIRCRAFT CATEGORY		DME THR DIST (m)	
2. OBSTACLE CLEARANCE OF BASIC ILS SURFACE				



CONTROLLING OBSTACLE (ID/NAME)	
OBSTACLE COORDINATION (LAT/LONG)	
OBSTACLE HEIGHT + TREE + ACCURACY (m)	
RC-X VALUE AT OBSTACLE LOCATION (m)	
RC-Y VALUE AT OBSTACLE LOCATION (m)	
OBSTACLE LOCATION	
SURFACE HEIGHT AT OBST LOCATION (m)	
PENETRATION (m) : Obstacle height - Surface Height Positive number=Penetration / Negative number=No Penetration	
HEIGHT LOSS MARGIN (m)	
COMPUTATION DH (ft) : MAX [OBST HGT + Height Loss / 200ft]	
COMPUTATION DA (ft) : DH + RWY ELEV	

3. OBSTACLE ASSESSMENT SURFACE(OAS) FOR CAT I

A. Category I/GP angle 3°/LOC-THR 3 000 m/missed approach gradient 2.5 per cent.



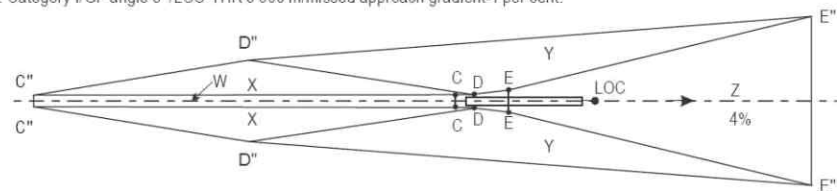
Equations of the obstacle assessment surfaces:

$$\begin{aligned} W I z &= 0.0285x - 8.01 \\ X I z &= 0.027681x + 0.1825y - 16.72 \\ Y I z &= 0.023948x + 0.210054y - 21.51 \\ Z I z &= -0.025x - 22.50 \end{aligned}$$

Coordinates of points C, D, E, C'', D'', E'', (m)

	C	D	E	C''	D''	E''
x	281	-286	-900	10 807	5 438	-12 900
y	49	135	205	96	910	3 001
z	0	0	0	300	300	300

B. Category I/GP angle 3°/LOC-THR 3 000 m/missed approach gradient 4 per cent.



Equations of the obstacle assessment surfaces:

$$\begin{aligned} W I z &= 0.0285x - 8.01 \\ X I z &= 0.027681x + 0.1825y - 16.72 \\ Y I z &= 0.020158x + 0.238021y - 26.37 \\ Z I z &= -0.04x - 36.00 \end{aligned}$$

Coordinates of points C, D, E, C'', D'', E'', (m)

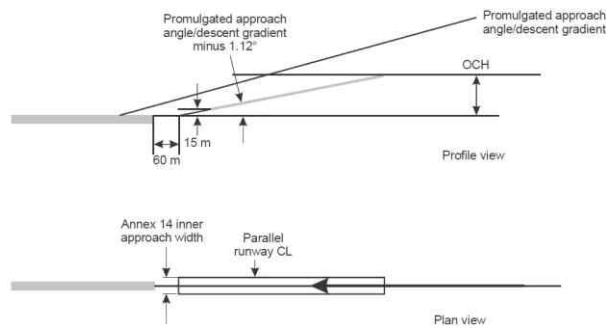
	C	D	E	C''	D''	E''
x	281	-286	-900	10 807	5 438	-8 400
y	49	135	187	96	910	2 082
z	0	0	0	300	300	300

3-1. OAS DATA							
APPROACH CATEGORY		GLIDE PATH(°)	LOC THR DIST(m)		RDH(m)	COURSE WIDTH at THR(m)	
MISSED APP CG(%)		AIRCRAFT CATEGORY			STANDARD		
3-2. FAF or FAP DATA							
DISTANCE OF FAF or FAP (NM) : From DME							
FAF or FAP ALTITUDE (m)							
RWY ELEVATION (m)							
FAF or FAP HEIGHT (m)							
INTERMEDIATE MOC (m)							
OAS HEIGHT (m) : FAP - IMT MOC							
3-3. OAS CONSTANTS & TEMPLATE COORDINATES : OAS HEIGHT is 300m							
FORMULA		A		B		C	
W Surface	Ax+C						
X Surface	Ax+By+C						
Y Surface	Ax+By+C						
Z Surface	Ax+C						
	C*	D*	C	D	E	E**	
X							
Y							
Z							
3-4. OAS CONSTANTS & TEMPLATE COORDINATES : OAS HEIGHT is more than 300m							
FORMULA		A		B		C	
W Surface	Ax+C						
X Surface	Ax+By+C						
Y Surface	Ax+By+C						
Z Surface	Ax+C						
	C*	D*	C	D	E	E**	
X							
Y							
Z							
3-5. OAS CONSTANTS & TEMPLATE COORDINATES : Use FAF (descent fix)							
FORMULA		A		B		C	
W Surface	Ax+C						
X Surface	Ax+By+C						
Y Surface	Ax+By+C						
Z Surface	Ax+C						
	FAF+TOL	C*	D*	C	D	E	E**
X							
Y							
Z							
3-6. OBSTACLE CLEARANCE OF OAS - APPROACH PART							

CONTROLLING OBSTACLE (ID/NAME)	
OBSTACLE COORDINATION (LAT/LONG)	
OBSTACLE HEIGHT + TREE + ACCURACY (m)	
RC-X VALUE AT OBSTACLE LOCATION (m)	
RC-Y VALUE AT OBSTACLE LOCATION (m)	
OBSTACLE LOCATION	
SURFACE HEIGHT AT OBST LOCATION (m)	
PENETRATION (m) : Obstacle height - Surface Height Positive number=Penetration / Negative number=No Penetration	
HEIGHT LOSS MARGIN (m)	
COMPUTATION DH (ft) : MAX [OBST HGT + Height Loss / 200ft]	
COMPUTATION DA (ft) : DH + RWY ELEV	
3-7. OBSTACLE CLEARANCE OF OAS - MISSED APPROACH PART	
CONTROLLING OBSTACLE (ID/NAME)	
OBSTACLE COORDINATION (LAT/LONG)	
OBSTACLE HEIGHT + TREE + ACCURACY (m)	
RC-X VALUE AT OBSTACLE LOCATION (m)	
RC-Y VALUE AT OBSTACLE LOCATION (m)	
OBSTACLE LOCATION	
SURFACE HEIGHT AT OBST LOCATION (m)	
PENETRATION (m) : Obstacle height - Surface Height Positive number=Penetration / Negative number=No Penetration	
Z SURFACE HEIGHT (m) : If Obstacle located in Y surface	
Hma (m) Obstacle located in Z surface : Obstacle Height, Obstacle located in Y surface : Z surface Height+Penetration	
EQUIVALENT Ha BY OBSTACLE (m) : $h_a = \frac{h_{ma} \cot Z + (x_y + x)}{\cot Z + \cot \theta}$	
HEIGHT LOSS MARGIN (m)	
COMPUTATION DH (ft) : MAX [Ha + Height Loss / 200ft]	
COMPUTATION DA (ft) : DH + RWY ELEV	
APPROACH DA (ft)	
ADJUSTED DA (ft)	
4. CRM RESULT	
CONTROLLING OBSTACLE (ID/NAME)	
OBSTACLE COORDINATION (LAT/LONG)	
OBSTACLE ELEVATION + TREE + ACCURACY (m)	
RC-X VALUE AT OBSTACLE LOCATION (m)	
RC-Y VALUE AT OBSTACLE LOCATION (m)	
INDIVIDUAL RISK (CATEGORY D/DL)	
MINIMUM DA (ft)	
ADJUSTED DA (ft)	
5. TEMPORARY DA OF PRECISION SEGMENT	
DA OF BASIC ILS SURFACE (ft)	
DA OF OAS (ft)	
DA OF CRM (ft)	
TEMPORARY DA OF PRECISION SEGMENT (ft)	
6. OBSTACLE CLEARANCE OF MISSED APPROACH - After OAS	

CONTROLLING OBSTACLE (ID/NAME)	
OBSTACLE COORDINATION (LAT/LONG)	
OBSTACLE ELEVATION + TREE + ACCURACY (ft)	
OBSTACLE LOCATION-1 (STRAIGHT / TURNING)	
OBSTACLE LOCATION-2 (P or S)	
MOC (ft)	
do VALUE (ft) : From SOC to OBST Location	
tanZ VALUE (%)	
APPROACH DA (ft)	
Height Loss (ft)	
STRAIGHT : (DA - HL) + dotanZ (ft) TURNING : (DA - HL) + dotanZ - MOC (ft)	
PENETRATION (ft) OBST ALT - [(DA - HL) + dotanZ] (STRAIGHT) OBST ALT - [(DA - HL) + dotanZ - MOC] (TURNING) Positive number=Penetration / Negative number=No Penetration	
ADJUSTED DA (ft) : 2.5% Climb gradient	
PUBLISHED DA/H (ft) FOR 2.5% CG	

7. VSS (VISUAL SEGMENT SURFACE)



7-1. VSS DIMENSIONS

Descent Angle (°)			RC-X	RC-Y(+)	RC-Y(-)	Z(m)
Angle of VSS (°)		BEGIN (ORIGIN)				
OCH (ft)		END (TERMINATION)				

7-2. OBSTACLE ASSESSMENT

CONTROLLING OBSTACLE (ID/NAME)	
OBSTACLE COORDINATION (LAT/LONG)	
OBSTACLE ELEVATION + TREE + ACCURACY (m)	
RC-X VALUE AT OBSTACLE LOCATION (m)	
RC-Y VALUE AT OBSTACLE LOCATION (m)	
SURFACE ELEVATION AT OBST (m) : Surface HGT + THR ELEV	
PENETRATION (m) Positive number=Penetration / Negative number=No Penetration	

8. VISIBILITY WITHOUT APPROACH LIGHTS

DH (ft)					
APPROACH CATEGORY	A	B	C	D	DL
REQUIRED VIS FOR DH TO RWY THR DISTANCE (m)					
VISIBILITY FROM ICAO DOC 9365 TABLE 6-1 (m)					
PUBLISHED VISIBILITY (m) : Highest from above					

9. VISIBILITY APPROACH LIGHTS

TYPE & LENGTH OF APPROACH LIGHTS	
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DH (ft)							
APPROACH CATEGORY			A	B	C	D	DL
LIGHT REDUCTION (m) : FALS : 720m, IALS : 420m, BALS : 210m							
VISIBILITY FROM ICAO DOC 9365 TABLE 6-1 (m)							
VISIBILITY (m) : Highest from above							
10. PUBLISHED MINIMA							
CATEGORY		DA(DH)	A	B	C	D	DL
CAT I (CG 2.5%)	FULL						
	ALS INOP						
11. REMARK							

2. ILS APPROACH PROCEDURE(CAT II & III)

 국토교통부 Ministry of Land, Infrastructure and Transport		ILS APPROACH PROCEDURE					
1. TITLE OF PROCEDURE		2. RWY NUMBER			3. REVISION DATE		
ILS Z RWY 07 CAT II							
4. OPERATING AGENCY		5. NEW OR REVISED			6. APPROVAL DATE		
7. AD INFORMATION							
AD NAME	ICAO IDENTIFIER	AD ELEV (ft)		ARP LAT/LONG		ARP MAG VAR	
8. USE RWY INFORMATION							
RWY NO	DIMENSION (m)	LAT/LONG		THR ELEV (ft)		BEARING(T°/M°)	
9. VISUAL AIDS							
APPROACH LIGHTS		VASIS/PAPI		RUNWAY LIGHTS		OTHER LIGHTS	
NOTE							
10. USE NAVAID INFORMATION							
ILS RDH (ft)			GP ANGLE (°)				
NAME	LAT/LONG		ELEV (ft)		MAG VAR		
11. SPECIFICATION & USE SENSOR							
SPECIFICATION	GNSS		DME/DME		VOR/DME		
12. MAP REFERENCE							
PUBLISHING OFFICE	DATE OF PUBLICATION		MAP NAME		SCALE		
13. PROCEDURE & REVISION SUMMARY							
14. PLAN VIEW OPERATIONAL NOTES							
15. MISSED APPROACH INSTRUCTION							
16. LANDING MINIMA							
CATEGORY		DA (DH)	A	B	C	D	DL
CAT I	FULL						

		ALS INOP								
CAT II										
NOTE										
17. FIX DATA										
TYPE		IDENT		WP/FIX		LAT/LONG				
IAF										
IF										
FAP										
THR										
MATF										
MAHF										
18. TRACK & SEGMENT DATA										
TYPE	IDENT	TRK °M (°T)	DIST (NM)	ALT (ft)	DG (%)		SPEED (kt)			
IAF										
IF										
FAP										
THR										
MATF										
MAHF										
NOTE										
19. MINIMUM STABLIZATION DISTANCE(MSD) - RNAV SEGMENTS ONLY										
SEGMENT		DESIGN DIST(NM)		MSD REQUIRED(NM)		REMAIN(NM)		CONFIRMATION		
20. TURN PARAMETER										
TYPE	IDENT	TURN ANLGE (°)	RADIUS OF TURN[r] (NM)	WIND EFFECT [E90°] (NM)	EARLIEST TURN POINT (NM)		LATEST TURN POINT (NM)			
21. CONTROLLING OBSTACLES										
SEGMENT		ID	NAME	CODE	TYPE	ALTITUDE (ft)		LAT/LONG		
Initial										
Intermediate										
Final1)										
Missed APP (Straight)										
Missed APP (Turning)										
VSS										
NOTE										
22. HOLDING INFORMATION										
IDENT	DIRECTION	ID	NAME	LAT/LONG	ELEV (ft)	MOC (ft)	MIN/MAX	SPEED (kt)		

							ALTITUDE (ft)	
23. MINIMUM SECTOR ALTITUDE (MSA)								
NAVAID/WPT			TYPE			IDENT		
TO	FROM	ARC (NM)	ID	NAME	ALT (ft)	LAT/LONG	MOC (ft)	MSA (ft)
NOTE								
24. EXEMPTION CRITERIA								
REQUIRED		REASON	-					
NOTE								
25. FLIGHT CHECK INFORMATION								
DATE OF FLIGHT CHECK		REASON					REMARKS	
26. CONCLUSION								
27. DEVELOPED BY								
NAME					DATE			
28. APPROVED BY								
NAME					DATE			
29. REMARKS								

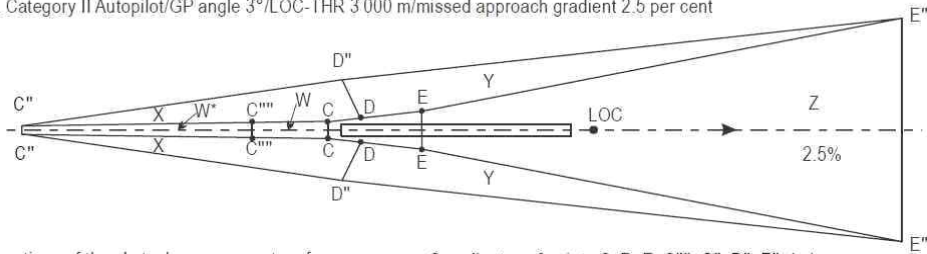
	AERONAUTICAL DATA TABULATION	
ILS/LOC Approach to RWY 07 from YUMIN(IAF)		
Fix/point		Coordinates

	PRECISION COMPUTATIONS for CAT II & III			
1. COMPUTATION DATA				
THR ELEV(m)	ILS RDH(m)	LOC THR DIST(m)	GP ANGLE(°)	COURSE WIDTH at THR(m)

MISSED APP CG(%)	AIRCRAFT CATEGORY	DME THR DIST (m)

2. OBSTACLE ASSESSMENT SURFACE(OAS) FOR CAT II

C. Category II Autopilot/GP angle 3°/LOC-THR 3 000 m/missed approach gradient 2.5 per cent



Equations of the obstacle assessment surfaces:

$$\begin{aligned} W \text{ I } z &= 0.0358x - 6.19 \\ W^* \text{ I } z &= 0.042x - 12.39 \\ X \text{ I } z &= 0.041370x + 0.2752y - 25.32 \\ Y \text{ I } z &= 0.031955x + 0.280291y - 28.70 \\ Z \text{ I } z &= -0.025x - 22.50 \end{aligned}$$

Coordinates of points C, D, E, C'', D'', E'', (m)

	C	D	E	C''	C''	D''	E''
x	173	-286	-900	1 000	3 866	-13	-6 900
y	66	135	205	49	55	639	1 424
z	0	0	0	29.6	150	150	150

2-1. OAS DATA

APPROACH CATEGORY	GLIDE PATH(°)	LOC THR DIST(m)	RDH(m)	COURSE WIDTH at THR(m)

MISSED APP CG(%)	AIRCRAFT CATEGORY	STANDARD

2-2. FAF or FAP DATA

DISTANCE OF FAF or FAP (NM) : From DME	
FAF or FAP ALTITUDE (m)	
RWY ELEVATION (m)	
FAF or FAP HEIGHT (m)	
INTERMEDIATE MOC (m)	
OAS HEIGHT (m) : FAP - IMT MOC	

2-3. OAS CONSTANTS & TEMPLATE COORDINATES

FORMULA	A	B	C
W Surface	Ax+C		
X Surface	Ax+By+C		
Y Surface	Ax+By+C		
Z Surface	Ax+C		

	C''	D''	C	D	E	E''
X						
Y						
Z						

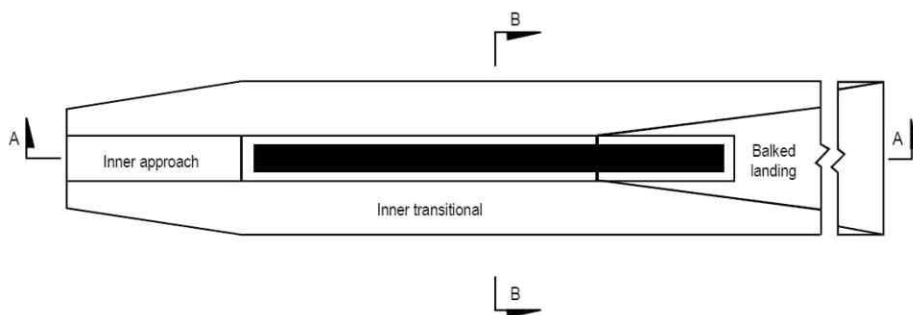
2-4. OBSTACLE CLEARANCE OF OAS - APPROACH PART

CONTROLLING OBSTACLE (ID/NAME)	
OBSTACLE COORDINATION (LAT/LONG)	
OBSTACLE HEIGHT + TREE + ACCURACY (m)	
RC-X VALUE AT OBSTACLE LOCATION (m)	
RC-Y VALUE AT OBSTACLE LOCATION (m)	
OBSTACLE LOCATION	
SURFACE HEIGHT AT OBST LOCATION (m)	

PENETRATION (m) : Obstacle height - Surface Height Positive number=Penetration / Negative number=No Penetration	
HEIGHT LOSS MARGIN (m)	
COMPUTATION DH (ft) : MAX [OBST HGT + Height Loss / 200ft]	
COMPUTATION DA (ft) : DH + RWY ELEV	
2-5. OBSTACLE CLEARANCE OF OAS - MISSED APPROACH PART	
CONTROLLING OBSTACLE (ID/NAME)	
OBSTACLE COORDINATION (LAT/LONG)	
OBSTACLE HEIGHT + TREE + ACCURACY (m)	
RC-X VALUE AT OBSTACLE LOCATION (m)	
RC-Y VALUE AT OBSTACLE LOCATION (m)	
OBSTACLE LOCATION	
SURFACE HEIGHT AT OBST LOCATION (m)	
PENETRATION (m) : Obstacle height - Surface Height Positive number=Penetration / Negative number=No Penetration	
Z SURFACE HEIGHT (m) : If Obstacle located in Y surface	
Hma (m) Obstacle located in Z surface : Obstacle Height, Obstacle located in Y surface : Z surface Height+Penetration	
EQUIVALENT Ha BY OBSTACLE (m) :	$h_a = \frac{h_{ma} \cot Z + (x_y + x)}{\cot Z + \cot \theta}$
HEIGHT LOSS MARGIN (m)	
COMPUTATION DH (ft) : MAX [Ha + Height Loss / 200ft]	
COMPUTATION DA (ft) : DH + RWY ELEV	
APPROACH DA (ft)	
ADJUSTED DA (ft)	
3. CRM RESULT	
CONTROLLING OBSTACLE (ID/NAME)	
OBSTACLE COORDINATION (LAT/LONG)	
OBSTACLE ELEVATION + TREE + ACCURACY (m)	
RC-X VALUE AT OBSTACLE LOCATION (m)	
RC-Y VALUE AT OBSTACLE LOCATION (m)	
INDIVIDUAL RISK (CATEGORY D/DL)	
MINIMUM DA (ft)	
ADJUSTED DA (ft)	
4. TEMPORARY DA OF PRECISION SEGMENT	
DA OF OAS (ft)	
DA OF CRM (ft)	
TEMPORARY DA OF PRECISION SEGMENT (ft)	
5. OBSTACLE CLEARANCE OF MISSED APPROACH - After OAS	
CONTROLLING OBSTACLE (ID/NAME)	
OBSTACLE COORDINATION (LAT/LONG)	
OBSTACLE ELEVATION + TREE + ACCURACY (ft)	
OBSTACLE LOCATION-1 (STRAIGHT / TURNING)	
OBSTACLE LOCATION-2 (P or S)	
MOC (ft)	
do VALUE (ft) : From SOC to OBST Location	

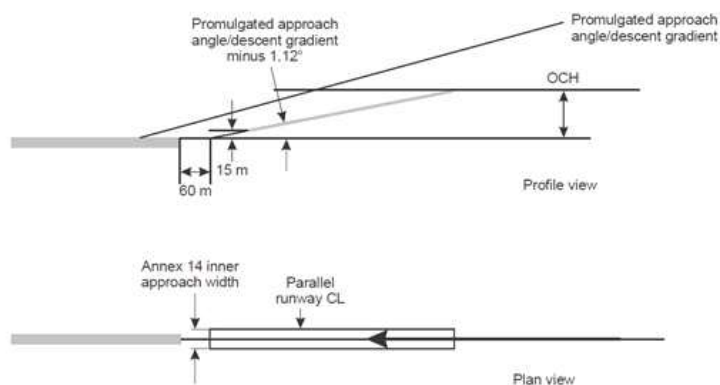
tanZ VALUE (%)	
APPROACH DA (ft)	
Height Loss (ft)	
STRAIGHT : (DA - HL) + dotanZ (ft) TURNING : (DA - HL) + dotanZ - MOC (ft)	
PENETRATION (ft) OBST ALT - [(DA - HL) + dotanZ] (STARIGHT) OBST ALT - [(DA - HL) + dotanZ - MOC] (TURNING) Positive number=Penetration / Negative number=No Penetration	
ADJUSTED DA (ft) : 2.5% Climb gradient	
PUBLISHED DA/H (ft) FOR 2.5% CG	

6. ANNEX 14 OBSTACLE ANALYSIS



CONTROLLING OBSTACLE (ID/NAME)	
OBSTACLE COORDINATION (LAT/LONG)	
OBSTACLE HEIGHT+ TREE + ACCURACY (m)	
RC-X VALUE AT OBSTACLE LOCATION (m)	
RC-Y VALUE AT OBSTACLE LOCATION (m)	
OBSTACLE LOCATION	
SURFACE HEIGHT (m)	
PENETRATION (m) : OBST HGT - Surface HGT Positive number=Penetration / Negative number=No Penetration	

7. VSS (VISUAL SEGMENT SURFACE)



7-1. VSS DIMENSIONS

Descent Angle (°)			RC-X	RC-Y(+)	RC-Y(-)	Z(m)
Angle of VSS (°)		BEGIN (ORIGIN)				
OCH (ft)		END (TERMINATION)				

7-2. OBSTACLE ASSESSMENT

CONTROLLING OBSTACLE (ID/NAME)	
OBSTACLE COORDINATION (LAT/LONG)	
OBSTACLE ELEVATION + TREE + ACCURACY (m)	
RC-X VALUE AT OBSTACLE LOCATION (m)	

RC-Y VALUE AT OBSTACLE LOCATION (m)						
SURFACE ELEVATION AT OBST (m) : Surface HGT + THR ELEV						
PENETRATION (m) Positive number=Penetration / Negative number=No Penetration						
8. VISIBILITY APPROACH LIGHTS (CAT II)						
DH (ft)						
APPROACH CATEGORY		A	B	C	D	DL
VISIBILITY FROM ICAO DOC 9365 TABLE 6-4 (m)						
9. VISIBILITY APPROACH LIGHTS (CAT III)						
DH (ft)						
APPROACH CATEGORY		A	B	C	D	DL
VISIBILITY FROM ICAO DOC 9365 TABLE 6-5 (m)						
10. PUBLISHED MINIMA						
CATEGORY	DA(DH)	A	B	C	D	DL
CAT II						
CAT III						
11. REMARKS						

3. LOC APPROACH PROCEDURE

 국토교통부 Ministry of Land, Infrastructure and Transport		LOC APPROACH PROCEDURE							
1. TITLE OF PROCEDURE		2. RWY NUMBER			3. REVISION DATE				
LOC Z RWY 07									
4. OPERATING AGENCY		5. NEW OR REVISED			6. APPROVAL DATE				
7. AD INFORMATION									
AD NAME		ICAO IDENTIFIER		AD ELEV (ft)		ARP LAT/LONG		ARP MAG VAR	
8. USE RWY INFORMATION									
RWY NO		DIMENSION (m)		LAT/LONG		THR ELEV (ft)		BEARING(T°/M°)	
9. VISUAL AIDS									
APPROACH LIGHTS		VASIS/PAPI		RUNWAY LIGHTS		OTHER LIGHTS			
NOTE									
10. USE NAVAID INFORMATION									
ILS RDH (ft)				GP ANGLE (°)					
NAME		LAT/LONG		ELEV (ft)		MAG VAR			
11. SPECIFICATION & USE SENSOR									
SPECIFICATION		GNSS		DME/DME		VOR/DME			
12. MAP REFERENCE									
PUBLISHING OFFICE		DATE OF PUBLICATION		MAP NAME		SCALE			
13. PROCEDURE & REVISION SUMMARY									
14. PLAN VIEW OPERATIONAL NOTES									
15. MISSED APPROACH INSTRUCTION									
16. LANDING MINIMA									
CATEGORY		DA (DH)	A	B	C	D	DL		
LOC		FULL							
		ALS INOP							

NOTE						
17. RECOMMENDED PROFILE						
		DME YDM				
		ALT(HGT)				
18. FIX DATA						
TYPE	IDENT	WP/FIX	LAT/LONG			
IAF						
IF						
FAF						
SDF						
MAPt						
THR						
MATF						
MAHF						
19. TRACK & SEGMENT DATA						
TYPE	IDENT	TRK °M (°T)	DIST (NM)	ALT (ft)	DG (%)	SPEED (kt)
IAF						
IF						
FAF						
SDF						
MAPt						
THR						
MATF						
MAHF						
NOTE						
20. MINIMUM STABILIZATION DISTANCE(MSD) - RNAV SEGMENTS ONLY						
SEGMENT	DESIGN DIST(NM)	MSD REQUIRED(NM)	REMAIN(NM)	CONFIRMATION		
21. TURN PARAMETER						
TYPE	IDENT	TURN ANLGE (°)	RADIUS OF TURN[r] (NM)	WIND EFFECT [E90°] (NM)	EARLIEST TURN POINT (NM)	LATEST TURN POINT (NM)
22. CONTROLLING OBSTACLES						
SEGMENT	ID	NAME	CODE	TYPE	ALTITUDE (ft)	LAT/LONG
Initial						
Intermediate						

Final (Before SDF)									
Final (After SDF)									
Missed APP (Straight)									
Missed APP (Turning)									
VSS									
NOTE									
23. HOLDING INFORMATION									
IDENT	DIRECTION	ID	NAME	LAT/LONG	ELEV (ft)	MOC (ft)	MIN/MAX ALTITUDE (ft)	SPEED (kt)	
24. MINIMUM SECTOR ALTITUDE (MSA)									
NAVAID/WPT				TYPE				IDENT	
TO	FROM	ARC (NM)	ID	NAME	ALT (ft)	LAT/LONG	MOC (ft)	MSA (ft)	
NOTE									
25. EXEMPTION CRITERIA									
REQUIRED		REASON	-						
NOTE									
26. FLIGHT CHECK INFORMATION									
DATE OF FLIGHT CHECK	REASON					REMARKS			
27. CONCLUSION									
28. DEVELOPED BY									
NAME				DATE					
29. APPROVED BY									
NAME				DATE					
30. REMARKS									

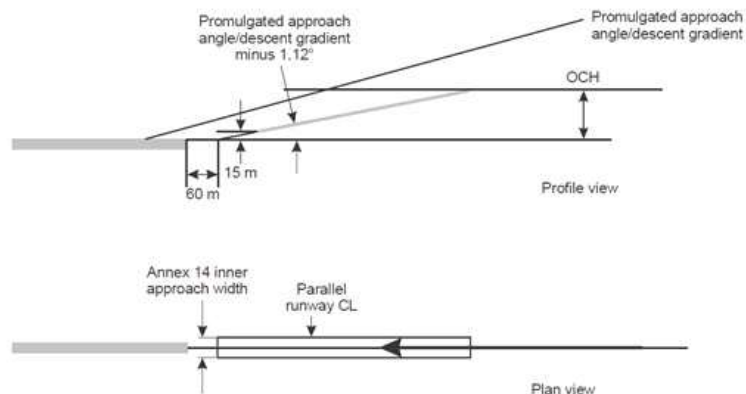
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		AERONAUTICAL DATA TABULATION
ILS/LOC Approach to RWY 07 from YUMIN(IAF)		
Fix/point		Coordinates

		LOC APPROACH COMPUTATIONS	
NAME OF AIRPORT		INSTRUMENT PROCEDURE	
1. MDA BASED ON FINAL APPROACH SEGMENT			
TYPE OF APPROACH			
FINAL APPROACH CONTROLLING OBSTACLE (ID/NAME)			
OBSTACLE (ft)			
LOCATION (P or S)			
MOC (ft)			
MDA ADJUSTMENTS (ft)			
UNCORRECT ADJUSTED MDA (ft)			
CORRECTED MDA (ft)			
2. MDA BASED ON MISSED APPROACH SEGMENT			
CORRECTED MDA (ft)			
CLIMB GRADIENT (%)			
CONTROLLING OBSTACLE (ID/NAME)			
OBSTACLE ELEVATION (ft)			
OBSTACLE LOCATION-1 : Initial, Intermediate, Final Pahse			
OBSTACLE LOCATION (P or S)			
DISTANCE FROM SOC TO OBST LOCATION (ft) : do			
ESTIMATED AIRCRAFT ALTITUDE AT OBST LOCATION (ft) : $MDA + do \tan Z$			
MOC AT OBSTACLE LOCATION (ft)			
ELEVATION OF SURFACE AT OBSTACLE (ft) : $MDA \text{ or } A/C \text{ ALT} - MOC$			
PENETRATION (ft) : $OBST \text{ ELEV} - SURFACE \text{ ELEV}$ Positive number=Penetration / Negative number=No Penetration			
IF PENETRATION EXISTS, ADJUSTED MDA AND/OR ESTABLISH CLIMB GRADIENT			
3. ADJUSTED MDA IF PENETRATION EXISTS			
AMOUNT OF PENETRATION TO BE ADJUSTED (ft)		-	
CORRECTED MDA (ft) : From Item 1			
UNCORRECTED ADJUSTED MDA (ft)			
CORRECTED ADJUSTED MDA (ft)			
4. MISSED APPROACH CLIMB GRADIENT IF PENETRATION EXISTS			
AMOUNT OF PENETRATION TO BE ADJUSTED (ft)		-	

CORRECTED MDA (ft) : From Item 1	
MOC (ft)	
DISTANCE FROM SOC TO OBSTACLE (ft) : do	
REQUIRED CLIMB GRADIENT (%) (OBST ELEV + MOC - MDA) / DISTANCE	
PUBLISHED CLIMB GRADIENT	

5. VSS (VISUAL SEGMENT SURFACE)



5-1. VSS DIMENSIONS

Descent Angle (°)			RC-X	RC-Y(+)	RC-Y(-)	Z(m)
Angle of VSS (°)		BEGIN (ORIGIN)				
OCH (ft)		END (TERMINATION)				

5-2. OBSTACLE ASSESSMENT

CONTROLLING OBSTACLE (ID/NAME)	
OBSTACLE COORDINATION (LAT/LONG)	
OBSTACLE ELEVATION + TREE + ACCURACY (m)	
RC-X VALUE AT OBSTACLE LOCATION (m)	
RC-Y VALUE AT OBSTACLE LOCATION (m)	
SURFACE ELEVATION AT OBST (m) : Surface HGT + THR ELEV	
PENETRATION (m) Positive number=Penetration / Negative number=No Penetration	

6. VISIBILITY WITHOUT APPROACH LIGHTS

MDH (ft)					
APPROACH CATEGORY	A	B	C	D	DL
REQUIRED VIS FOR MAPt TO RWY THR DISTANCE (m)					
VISIBILITY FROM ICAO DOC 9365 TABLE 6-1 (m)					
PUBLISHED VISIBILITY (m) : Highest from above					

7. VISIBILITY WITH APPROACH LIGHTS

TYPE & LENGTH OF APPROACH LIGHTS					
MDH (ft)					
APPROACH CATEGORY	A	B	C	D	DLIGHT REDUCTION (m) : FALS : 720m, IALS : 420m, BALS : 210m160016 0016001600

4. VOR APPROACH PROCEDURE

 국토교통부 Ministry of Land, Infrastructure and Transport		VOR APPROACH PROCEDURE					
1. TITLE OF PROCEDURE		2. RWY NUMBER			3. REVISION DATE		
VOR RWY 07							
4. OPERATING AGENCY		5. NEW OR REVISED			6. APPROVAL DATE		
7. AD INFORMATION							
AD NAME	ICAO IDENTIFIER	AD ELEV (ft)	ARP LAT/LONG	ARP MAG VAR			
8. USE RWY INFORMATION							
RWY NO	DIMENSION (m)	LAT/LONG	THR ELEV (ft)	BEARING(T°/M°)			
9. VISUAL AIDS							
APPROACH LIGHTS	VASIS/PAPI	RUNWAY LIGHTS		OTHER LIGHTS			
NOTE							
10. USE NAVAID INFORMATION							
RDH (ft)		GP ANGLE (°)					
NAME	LAT/LONG	ELEV (ft)		MAG VAR			
11. SPECIFICATION & USE SENSOR							
SPECIFICATION	GNSS	DME/DME		VOR/DME			
12. MAP REFERENCE							
PUBLISHING OFFICE	DATE OF PUBLICATION	MAP NAME		SCALE			
13. PROCEDURE & REVISION SUMMARY							
14. PLAN VIEW OPERATIONAL NOTES							
15. MISSED APPROACH INSTRUCTION							
16. LANDING MINIMA							
CATEGORY		DA (DH)	A	B	C	D	DL
VOR	FULL						
	ALS INOP						
NOTE							

17. RECOMMENDED PROFILE

Final Approach Gradient 5.34%, 325ft/NM	DME YDM				
	ALT(HGT)				

18. FIX DATA

TYPE	IDENT	WP/FIX	LAT/LONG

19. TRACK & SEGMENT DATA

TYPE	IDENT	TRK °M (°T)	DiST (NM)	ALT (ft)	DG (%)	SPEED (kt)

[illegible]

20. MINIMUM STABILIZATION DISTANCE(MSD) - RNAV SEGMENTS ONLY

SEGMENT	DESIGN DIST(NM)	MSD REQUIRED(NM)	REMAIN(NM)	CONFIRMATION

21. TURN PARAMETER

TYPE	IDENT	TURN ANLGE (°)	RADIUS OF TURN[r] (NM)	WIND EFFECT [E90°] (NM)	EARLIEST TURN POINT (NM)	LATEST TURN POINT (NM)
MATF						

22. CONTROLLING OBSTACLES

Segment	ID	Name	Code	Type	Altitude (ft)	Lat/Long
Initial						
Intermediate						
Final						
Missed APP (Straight)						
Missed APP (Turning)						
VSS						

[illegible]

23. HOLDING INFORMATION

IDENT	DIRECTION	ID	NAME	LAT/LONG	ELEV (ft)	MOC (ft)	MIN/MAX ALTITUDE (ft)	SPEED (kt)

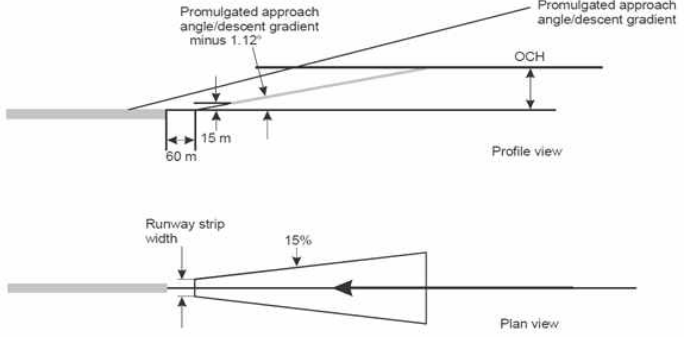
24. MINIMUM SECTOR ALTITUDE (MSA)

[illegible]

NOTE								
25. EXEMPTION CRITERIA								
REQUIRED		REASON						
NOTE								
26. FLIGHT CHECK INFORMATION								
DATE OF FLIGHT CHECK			REASON			REMARKS		
27. CONCLUSION								
28. DEVELOPED BY								
NAME					DATE			
29. APPROVED BY								
NAME					DATE			
30. REMARKS								

	AERONAUTICAL DATA TABULATION	
VOR Approach to RWY 07 from ANBIN (IAF)		
Fix/point		Coordinates

	VOR APPROACH COMPUTATIONS		
NAME OF AIRPORT		INSTRUMENT PROCEDURE	
1. MDA BASED ON FINAL APPROACH SEGMENT			
TYPE OF APPROACH			
FINAL APPROACH CONTROLLING OBSTACLE (ID/NAME)			
OBSTACLE (ft)			
LOCATION (P or S)			
MOC (ft)			
MDA ADJUSTMENTS (ft)			
UNCORRECT ADJUSTED MDA (ft)			
CORRECTED MDA (ft)			

2. MDA BASED ON MISSED APPROACH SEGMENT						
CORRECTED MDA (ft)						
CLIMB GRADIENT (%)						
CONTROLLING OBSTACLE (ID/NAME)						
OBSTACLE ELEVATION (ft)						
OBSTACLE LOCATION-1 : Initial, Intermediate, Final Pahse						
OBSTACLE LOCATION (P or S)						
DISTANCE FROM SOC TO OBST LOCATION (ft) : do						
ESTIMATED AIRCRAFT ALTITUDE AT OBST LOCATION (ft) : MDA + do tanZ						
MOC AT OBSTACLE LOCATION (ft)						
ELEVATION OF SURFACE AT OBSTACLE (ft) : MDA or A/C ALT - MOC						
PENETRATION (ft) : OBST ELEV - SURFACE ELEV Positive number=Penetration / Negative number=No Penetration						
IF PENETRATION EXISTS, ADJUSTED MDA AND/OR ESTABLISH CLIMB GRADIENT						
3. ADJUSTED MDA IF PENETRATION EXISTS						
AMOUNT OF PENETRATION TO BE ADJUSTED (ft)			-			
CORRECTED MDA (ft) : From Item 1						
UNCORRECTED ADJUSTED MDA (ft)						
CORRECTED ADJUSTED MDA (ft)						
4. MISSED APPROACH CLIMB GRADIENT IF PENETRATION EXISTS						
AMOUNT OF PENETRATION TO BE ADJUSTED (ft)			-			
CORRECTED MDA (ft) : From Item 1						
MOC (ft)						
DISTANCE FROM SOC TO OBSTACLE (ft) : do						
REQUIRED CLIMB GRADIENT (%) (OBST ELEV + MOC - MDA) / DISTANCE						
PUBLISHED CLIMB GRADIENT						
5. VSS (VISUAL SEGMENT SURFACE)						
						
5-1. VSS DIMENSIONS						
Descent Angle (°)			RC-X	RC-Y(+)	RC-Y(-)	Z(m)
Angle of VSS (°)		BEGIN (ORIGIN)				
OCH (ft)		END (TERMINATION)				
5-2. OBSTACLE ASSESSMENT						
CONTROLLING OBSTACLE (ID/NAME)						
OBSTACLE COORDINATION (LAT/LONG)						
OBSTACLE ELEVATION + TREE + ACCURACY (m)						

RC-X VALUE AT OBSTACLE LOCATION (m)								
RC-Y VALUE AT OBSTACLE LOCATION (m)								
SURFACE ELEVATION AT OBST (m) : Surface HGT + THR ELEV								
PENETRATION (m) Positive number=Penetration / Negative number=No Penetration								
6. VISIBILITY WITHOUT APPROACH LIGHTS								
MDH (ft)								
APPROACH CATEGORY		A	B	C	D	DL		
REQUIRED VIS FOR MAPt TO RWY THR DISTANCE (m)								
VISIBILITY FROM ICAO DOC 9365 TABLE 6-1 (m)								
PUBLISHED VISIBILITY (m) : Highest from above								
7. VISIBILITY APPROACH LIGHTS								
TYPE & LENGTH OF APPROACH LIGHTS								
MDH (ft)								
APPROACH CATEGORY		A	B	C	D	DL		
LIGHT REDUCTION (m) : FALS : 720m, IALS : 420m, BALS : 210m								
VISIBILITY FROM ICAO DOC 9365 TABLE 6-1 (m)								
PUBLISHED VISIBILITY (m) : Highest from above								
8. PUBLISHED MINIMA								
CATEGORY		MDA(MDH)		A	B	C	D	DL
VOR	FULL	860(773)						
	ALS INOP							
9. REMARKS								

5. LNAV APPROACH PROCEDURE

 국토교통부 Ministry of Land, Infrastructure and Transport		RNAV(GNSS) APPROACH PROCEDURE - LNAV					
1. TITLE OF PROCEDURE		2. RWY NUMBER			3. REVISION DATE		
RNAV(GNSS) RWY 07 - LNAV							
4. OPERATING AGENCY		5. NEW OR REVISED			6. APPROVAL DATE		
7. AD INFORMATION							
AD NAME	ICAO IDENTIFIER	AD ELEV (ft)	ARP LAT/LONG	ARP MAG VAR			
8. USE RWY INFORMATION							
RWY NO	DIMENSION (m)	LAT/LONG	THR ELEV (ft)	BEARING(T°/M°)			
9. VISUAL AIDS							
APPROACH LIGHTS	VASIS/PAPI	RUNWAY LIGHTS		OTHER LIGHTS			
NOTE							
10. SPECIFICATION & USE SENSOR							
LNAV RDH (ft)		VPA (°)					
SPECIFICATION	GNSS	DME/DME	VOR/DME				
11. MAP REFERENCE							
PUBLISHING OFFICE	DATE OF PUBLICATION	MAP NAME	SCALE				
12. PROCEDURE & REVISION SUMMARY							
13. PLAN VIEW OPERATIONAL NOTES							
14. MISSED APPROACH INSTRUCTION							
15. LANDING MINIMA							
CATEGORY		DA (DH)	A	B	C	D	DL
LNAV	FULL						
	ALS INOP						
NOTE							
16. RECOMMENDED PROFILE							
Final Approach Gradient 5.24%, 319ft/NM	FROM RWT						
	ALT(HGT)						
17. FIX DATA							

TYPE		IDENT		WP TYPE		LAT/LONG	

18. TRACK & SEGMENT DATA

TYPE		IDENT		TRK 'M ('T)		DiST (NM)		ALT (ft)		DG (%)		SPEED (kt)	

NOTE													
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19. MINIMUM STABILIZATION DISTANCE(MSD)

SEGMENT		DESIGN DIST(NM)		MSD REQUIRED(NM)		REMAIN(NM)		CONFIRMATION	

20. TURN PARAMETER

TYPE		IDENT		TURN ANLGE (°)		RADIUS OF TURN[r] (NM)		WIND EFFECT [E90°] (NM)		EARLIEST TURN POINT (NM)		LATEST TURN POINT (NM)	

21. CONTROLLING OBSTACLES

SEGMENT		ID		NAME		CODE		TYPE		ALTITUDE (ft)		LAT/LONG	
Initial													
Intermediate													
Final (Before SDF)													
Final (After SDF)													
Missed APP (Straight)													
Missed APP (Turning)													
VSS													

NOTE													
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22. HOLDING INFORMATION

IDENT		DIRECTION		ID		NAME		LAT/LONG		ELEV (ft)		MOC (ft)		MIN/MAX ALTITUDE (ft)		SPEED (kt)	

23. MINIMUM SECTOR ALTITUDE (MSA)

NAVAID/WPT				TYPE				IDENT									
TO		FROM		ARC (NM)		ID		NAME		ALT (ft)		LAT/LONG		MOC (ft)		MSA (ft)	

NOTE																	
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24. EXEMPTION CRITERIA			
REQUIRED		REASON	-
NOTE			
25. FLIGHT CHECK INFORMATION			
DATE OF FLIGHT CHECK	REASON		REMARKS
26. CONCLUSION			
27. DEVELOPED BY			
NAME		DATE	
28. APPROVED BY			
NAME		DATE	
29. REMARKS			

AERONAUTICAL DATA TABULATION												
RNAV(GNSS) RWY 07												
Serial Number	Path Descriptor	Waypoint Identifier	Fly-over	Course/Track °M(°T)	Distance (NM)	Turn Direction	Altitude (ft)	Speed (kt)	Coordinates	VPA/RDH	Navigation specification	Remarks
1												
2												
3												
HOLDING PROCEDURE												
Holding Identification	Path Descriptor	Waypoint Identifier	Fly-over	Course/Track °M(°T)	Time (min)	Turn Direction	Altitude (ft)	Speed (kt)	Coordinates	VPA/RDH	Navigation specification	Remarks

NONPRECISION COMPUTATIONS - LNAV	
NAME OF AIRPORT	INSTRUMENT PROCEDURE
1. MDA BASED ON FINAL APPROACH SEGMENT	
TYPE OF APPROACH	
FINAL APPROACH CONTROLLING OBSTACLE (ID/NAME)	
OBSTACLE (ft)	
LOCATION (P or S)	

MOC (ft)						
MDA ADJUSTMENTS (ft)						
UNCORRECT ADJUSTED MDA (ft)						
CORRECTED MDA (ft)						
2. MDA BASED ON MISSED APPROACH SEGMENT						
CORRECTED MDA (ft)						
CLIMB GRADIENT (%)						
CONTROLLING OBSTACLE (ID/NAME)						
OBSTACLE ELEVATION (ft)						
OBSTACLE LOCATION-1 : Initial, Intermediate, Final Pahse						
OBSTACLE LOCATION (P or S)						
DISTANCE FROM SOC TO OBST LOCATION (ft) : do						
ESTIMATED AIRCRAFT ALTITUDE AT OBST LOCATION (ft) : MDA + do tanZ						
MOC AT OBSTACLE LOCATION (ft)						
ELEVATION OF SURFACE AT OBSTACLE (ft) : MDA or A/C ALT - MOC						
PENETRATION (ft) : OBST ELEV - SURFACE ELEV Positive number=Penetration / Negative number=No Penetration						
3. ADJUSTED MDA IF PENETRATION EXISTS						
AMOUNT OF PENETRATION TO BE ADJUSTED (ft)	-					
CORRECTED MDA (ft) : From Item 1						
UNCORRECTED ADJUSTED MDA (ft)						
CORRECTED ADJUSTED MDA (ft)						
4. MISSED APPROACH CLIMB GRADIENT IF PENETRATION EXISTS						
AMOUNT OF PENETRATION TO BE ADJUSTED (ft)	-					
CORRECTED MDA (ft) : From Item 1						
MOC (ft)						
DISTANCE FROM SOC TO OBSTACLE (ft) : do						
REQUIRED CLIMB GRADIENT (%) (OBST ELEV + MOC - MDA) / DISTANCE						
PUBLISHED CLIMB GRADIENT						
5. VSS (VISUAL SEGMENT SURFACE)						
5-1. VSS DIMENSIONS						
Descent Angle (°)			RC-X	RC-Y(+)	RC-Y(-)	Z(m)
Angle of VSS (°)		BEGIN (ORIGIN)				
OCH (ft)		END (TERMINATION)				
5-2. OBSTACLE ASSESSMENT						
CONTROLLING OBSTACLE (ID/NAME)						
OBSTACLE COORDINATION (LAT/LONG)						
OBSTACLE ELEVATION + TREE + ACCURACY (m)						
RC-X VALUE AT OBSTACLE LOCATION (m)						

SURFACE ELEVATION AT OBST (m) : Surface HGT + THR ELEV								
PENETRATION (m) Positive number=Penetration / Negative number=No Penetration								
6. VISIBILITY WITHOUT APPROACH LIGHTS								
MDH (ft)								
APPROACH CATEGORY		A	B	C	D	DL		
REQUIRED VIS FOR MAPt TO RWY THR DISTANCE (m)								
VISIBILITY FROM ICAO DOC 9365 TABLE 6-1 (m)								
PUBLISHED VISIBILITY (m) : Highest from above								
7. VISIBILITY APPROACH LIGHTS								
TYPE & LENGTH OF APPROACH LIGHTS								
MDH (ft)								
APPROACH CATEGORY		A	B	C	D	DL		
LIGHT REDUCTION (m) : FALS : 720m, IALS : 420m, BALS : 210m								
VISIBILITY FROM ICAO DOC 9365 TABLE 6-1 (m)								
PUBLISHED VISIBILITY (m) : Highest from above								
8. PUBLISHED MINIMA								
CATEGORY		MDA(MDH)		A	B	C	D	DL
LNAV	FULL							
	ALS INOP							
9. REMARKS								

6. LNAV/VNAV APPROACH PROCEDURE

 국토교통부 Ministry of Land, Infrastructure and Transport		RNAV(GNSS) APPROACH PROCEDURE - LNAV/VNAV					
1. TITLE OF PROCEDURE		2. RWY NUMBER			3. REVISION DATE		
RNAV(GNSS) RWY 07 - LNAV/VNAV							
4. OPERATING AGENCY		5. NEW OR REVISED			6. APPROVAL DATE		
7. AD INFORMATION							
AD NAME	ICAO IDENTIFIER	AD ELEV (ft)	ARP LAT/LONG	ARP MAG VAR			
8. USE RWY INFORMATION							
RWY NO	DIMENSION (m)	LAT/LONG	THR ELEV (ft)	BEARING(T°/M°)			
9. VISUAL AIDS							
APPROACH LIGHTS	VASIS/PAPI	RUNWAY LIGHTS		OTHER LIGHTS			
NOTE							
10. SPECIFICATION & USE SENSOR							
LNAV/VNAV RDH (ft)		VPA (°)					
SPECIFICATION	GNSS	DME/DME	VOR/DME				
11. MAP REFERENCE							
PUBLISHING OFFICE	DATE OF PUBLICATION	MAP NAME	SCALE				
12. PROCEDURE & REVISION SUMMARY							
13. PLAN VIEW OPERATIONAL NOTES							
14. MISSED APPROACH INSTRUCTION							
15. LANDING MINIMA							
CATEGORY		DA (DH)	A	B	C	D	DL
LNAV/VNAV	FULL						
	ALS INOP						
NOTE							
16. FIX DATA							
TYPE	IDENT	WP TYPE	LAT/LONG				

17. TRACK & SEGMENT DATA																
TYPE		IDENT		TRK °M (°T)		DiST (NM)		ALT (ft)		DG (%)		SPEED (kt)				
NOTE																
18. MINIMUM STABILIZATION DISTANCE(MSD)																
SEGMENT		DESIGN DIST(NM)			MSD REQUIRED(NM)			REMAIN(NM)			CONFIRMATION					
19. TURN PARAMETER																
TYPE		IDENT		TURN ANLGE (°)		RADIUS OF TURN[r] (NM)		WIND EFFECT [E90°] (NM)		EARLIEST TURN POINT (NM)		LATEST TURN POINT (NM)				
20. CONTROLLING OBSTACLES																
SEGMENT		ID	NAME		CODE	TYPE		ALTITUDE (ft)		LAT/LONG						
Initial																
Intermediate																
Final (Before SDF)																
Missed APP (Straight)																
Missed APP (Turning)																
VSS																
NOTE																
21. TEMPERATURE CORRECTION																
LOW TEMP (°C)			MIN VPA (°)			HIGH TEMP (°C)			MAX VPA (°)							
YEAR											Average					
LOW TEMP OF COLDEST MONTH																
HIGH TEMP OF HOTTEST MONTH																
22. HOLDING INFORMATION																
IDENT		DIRECTION		ID	NAME		LAT/LONG		ELEV (ft)		MOC (ft)		MIN/MAX ALTITUDE (ft)		SPEED (kt)	
23. MINIMUM SECTOR ALTITUDE (MSA)																
NAVAID/WPT					TYPE						IDENT					
TO		FROM		ARC (NM)		ID	NAME		ALT (ft)		LAT/LONG		MOC (ft)		MSA (ft)	

NOTE								
24. EXEMPTION CRITERIA								
REQUIRED		REASON						
NOTE								
25. FLIGHT CHECK INFORMATION								
DATE OF FLIGHT CHECK				REASON			REMARKS	
26. CONCLUSION								
27. DEVELOPED BY								
NAME					DATE			
28. APPROVED BY								
NAME					DATE			
29. REMARKS								

				AERONAUTICAL DATA TABULATION								
RNAV(GNSS) RWY 07												
Serial Number	Path Descriptor	Waypoint Identifier	Fly-over	Course/Track °M(°T)	Distance (NM)	Turn Direction	Altitude (ft)	Speed (kt)	Coordinates	VPA/RDH	Navigation specification	Remarks
1												
2												
3												
HOLDING PROCEDURE												
Holding Identification	Path Descriptor	Waypoint Identifier	Fly-over	Course/Track °M(°T)	Time (min)	Turn Direction	Altitude (ft)	Speed (kt)	Coordinates	VPA/RDH	Navigation specification	Remarks

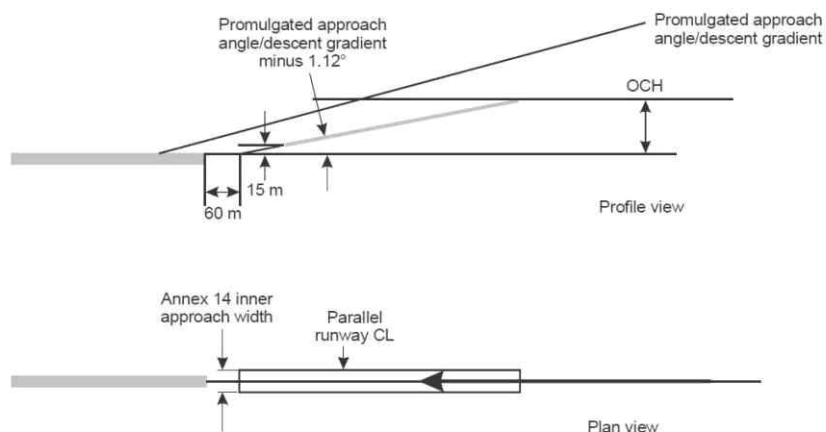
				NONPRECISION COMPUTATIONS - LNAV/VNAV			
NAME OF AIRPORT				INSTRUMENT PROCEDURE		LNAV/VNAV RWY 07	
1. DA BASED ON FINAL APPROACH SEGMENT							
TYPE OF APPROACH							
FINAL APPROACH CONTROLLING OBSTACLE (ID/NAME)							

OBSTACLE ELEVATION (ft)	
RWY THRESHOLD ELEVATION (ft)	
OBSTACLE HEIGHT (ft)m	
LOCATION-1	
LOCATION-2 (P or S)	
OAS HEIGHT AT OBSTACLE LOCATION (ft)	
PENETRATION (ft) : OBST ELEV - SURFACE ELEV Positive number=Penetration / Negative number=No Penetration	
HEIGHT LOSS MARGIN (ft)	
UNCORRECT ADJUSTED DH (ft) : OBST HIGT + HL	
MINIMUM DH (ft) : 246ft + HL [DOC 8168 VOL I]	
UNCORRECT ADJUSTED DA (ft)	
CORRECTED DA (ft)	
2. DA BASED ON MISSED APPROACH SEGMENT	
mMISSED APPROACH CONTROLLING OBSTACLE (ID/NAME)	
OBSTACLE ELEVATION (ft)	
RWY THRESHOLD ELEVATION (ft)	
OBSTACLE HEIGHT (ft)	
LOCATION-1	
LOCATION-2 (P or S)	
OAS HEIGHT OF PRIMARY AT OBSTACLE LOCATION (ft)	
OAS HEIGHT OF SECONDARY AT OBSTACLE LOCATION (ft)	
PENETRATION (ft) : OBST ELEV - SURFACE ELEV Positive number=Penetration / Negative number=No Penetration	
H'ma (ft) If OBST located in Secondary, OAS HGT of P + Penetration	
ha BY OBSTACLE (ft) :	
HEIGHT LOSS MARGIN (ft)	
UNCORRECT ADJUSTED DH (ft) : OBST HIGT + HL	
MINIMUM DH (ft) : 246ft + HL [DOC 8168 VOL I]	
UNCORRECT ADJUSTED DA (ft)	
CORRECTED DA (ft)	
TEMPERARY PUBLISHED DA (ft) Compare Final with Missed APP DA	
3. DA BASED ON MISSED APPROACH SEGMENT - AFTER OAS	
OBSTACLE (ID/NAME)	
OBSTACLE COORDINATION (LAT/LONG)	
OBSTACLE ELEVATION + TREE + ACCURACY (ft)	
OBSTACLE LOCATION-1 (STRAIGHT / TURNING)	
OBSTACLE LOCATION-2 (P or S)	
MOC (ft)	
do VALUE (ft) : From SOC to OBST Location	
tanZ. VALUE (%)	
APPROACH DA (ft)	
Height Loss (ft)	

STRAIGHT : $(DA - HL) + \text{dotan}Z$ (ft)	
TURNING : $(DA - HL) + \text{dotan}Z - \text{MOC}$ (ft)	
PENETRATION (ft) : $\text{OBST ALT} - [(DA - HL) + \text{dotan}Z]$ (STRAIGHT) : $\text{OBST ALT} - [(DA - HL) + \text{dotan}Z - \text{MOC}]$ (TURNING) Positive number=Penetration / Negative number=No Penetration	

IF PENETRATION EXISTS, ADJUSTED DA AND/OR ESTABLISH CLIMB GRADIENT

4. VSS (VISUAL SEGMENT SURFACE)



4-1. VSS DIMENSIONS

Descent Angle (°)			RC-X	RC-Y(+)	RC-Y(-)	Z(m)
Angle of VSS (°)		BEGIN (ORIGIN)				
OCH (ft)		END (TERMINATION)				

4-2. OBSTACLE ASSESSMENT

CONTROLLING OBSTACLE (ID/NAME)	
OBSTACLE COORDINATION (LAT/LONG)	
OBSTACLE ELEVATION + TREE + ACCURACY (m)	
RC-X VALUE AT OBSTACLE LOCATION (m)	
RC-Y VALUE AT OBSTACLE LOCATION (m)	
SURFACE ELEVATION AT OBST (m) : Surface HGT + THR ELEV	
PENETRATION (m) Positive number=Penetration / Negative number=No Penetration	

5. VISIBILITY WITHOUT APPROACH LIGHTS

MDH (ft)					
APPROACH CATEGORY	A	B	C	D	DL
REQUIRED VIS FOR MAPt TO RWY THR DISTANCE (m)					
VISIBILITY FROM ICAO DOC 9365 TABLE 6-1 (m)					
PUBLISHED VISIBILITY (m) : Highest from above					

6. VISIBILITY APPROACH LIGHTS

TYPE & LENGTH OF APPROACH LIGHTS					
MDH (ft)					
APPROACH CATEGORY	A	B	C	D	DL
LIGHT REDUCTION (m) : FALS : 720m, IALS : 420m, BALS : 210m					
VISIBILITY FROM ICAO DOC 9365 TABLE 6-1 (m)					
PUBLISHED VISIBILITY (m) : Highest from above					

7. PUBLISHED MINIMA

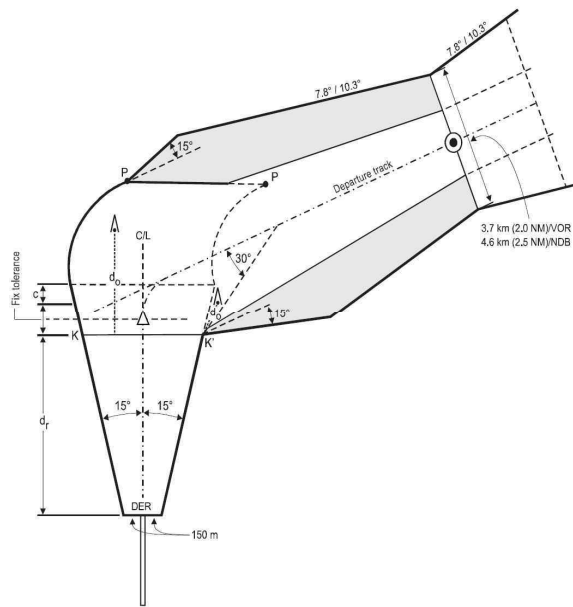
CATEGORY		MDA(MDH)	A	B	C	D	DL
LNAV/VNAV	FULL						
	ALS INOP						
8. REMARKS							

7. CONV SID(DEP)

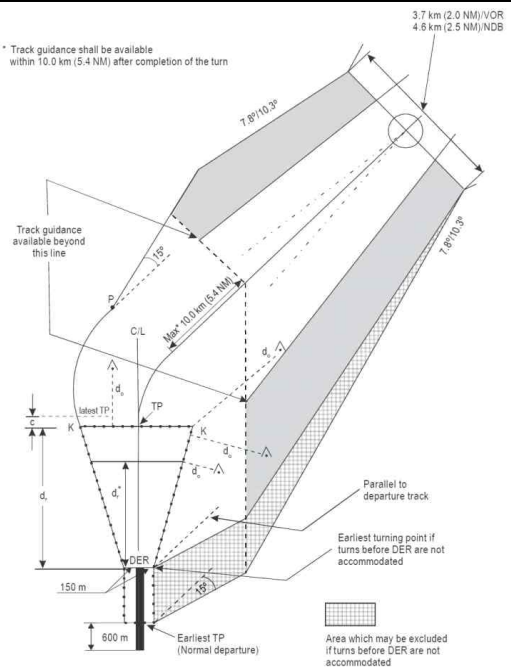
 국토교통부 Ministry of Land, Infrastructure and Transport		CONVENTIONAL DEPARTURE PROCEDURE			
1. TITLE OF PROCEDURE		2. RWY NUMBER		3. REVISION DATE	
4. OPERATING AGENCY		5. NEW OR REVISED		6. APPROVAL DATE	
7. AD INFORMATION					
AD NAME	ICAO IDENTIFIER	AD ELEV (ft)	ARP LAT/LONG	ARP MAG VAR	
8. USE RWY INFORMATION					
RWY NO	DIMENSION (m)	LAT/LONG	THR ELEV (ft)	BEARING(T°/M°)	
9. USE NAVAID INFORMATION					
NAME	LAT/LONG	ELEV (ft)	MAG VAR		
10. MAP REFERENCE					
PUBLISHING OFFICE	DATE OF PUBLICATION	MAP NAME	SCALE		
11. PROCEDURE & REVISION SUMMARY					
12. PLAN VIEW OPERATIONAL NOTES					
13. DEPARTURE ROUTE DESCRIPTION					
14. TRACK & SEGMENT DATA					
NAME	MAG TRK(°)	DISTANCE (NM)	ALT(ft)	SPEED(kt)	LAT/LONG
NOTE					
15. TURN PARAMETER					
IDENT	TURN ANLGE (°)	RADIUS OF TURN[r] (NM)	WIND EFFECT [E90°] (NM)	EARLIEST TURN POINT (NM)	LATEST TURN POINT (NM)
16. CONTROLLING OBSTACLES					
ID	NAME	CODE	TYPE	ALTITUDE (ft)	LAT/LONG

NOTE							
17. MINIMUM SECTOR ALTITUDE (MSA)							
NAVAID/WPT			TYPE			IDENT	
TO	FROM	ARC (NM)	ID	NAME	ALT (ft)	LAT/LONG	MOC (ft) MSA (ft)
NOTE							
18. EXEMPTION CRITERIA							
REQUIRED		REASON					
NOTE							
19. FLIGHT CHECK INFORMATION							
DATE OF FLIGHT CHECK		REASON				REMARKS	
20. CONCLUSION							
21. DEVELOPED BY							
NAME					DATE		
22. APPROVED BY							
NAME					DATE		
23. REMARKS							

		APPLICATION OF DEPARTURE PROCEDURE	
NAME OF AIRPORT			INSTRUMENT PROCEDURE
RWY NUMBER & DEPARTURE END OF RUNWAY ELEVATION			
1. EVALUATION RELATIONSHIPS			



* Track guidance shall be available within 10.0 km (5.4 NM) after completion of the turn



2. DEPARTURE ROUTE DESCRIPTION

3. OBSTACLE ANALYSIS BEFORE TURN POINT

CONTROLLING OBSTACLE (ID/NAME)	
OBSTACLE HEIGHT (ft)	
OIS HEIGHT AT DER (ft) :16ft	
DISTANCE FROM DER TO OBSTACLE (ft)	
MOC (ft) : Distance from DER to OBST * 0.8%	
REQUIRED HEIGHT AT OBST LOCATION (ft) : OBST HGT + MOC	
MINIMUM PDG FOR OBST AVOIDANCE (%) [OBST HGT + MOC - OIS Height at DER] / Distance	

4. OBSTACLE ANALYSIS AFTER TURN POINT

CONTROLLING OBSTACLE (ID/NAME)	
OBSTACLE HEIGHT (ft)	
OIS HEIGHT AT DER (ft) :16ft	
DISTANCE FROM DER TO KK' LINE : Dr (ft)	
SHORTEST DISTANCE FROM KK' LINE TO OBST : Do (ft)	
① (Dr+Do)*0.8% (ft) : MAX 1000ft (Mountainous area 2000ft)	
② MOC FOR TURNING (ft)	
SELECT MOC (ft) : MAX ①, ②	
REQUIRED HEIGHT AT OBST LOCATION (ft) : OBST HGT + MOC	
MINIMUM PDG FOR OBST AVOIDANCE (%) : [OBST HGT + MOC - OIS Height at DER] / Distance	

5. DETERMINE CLIMB GRADIENT AND ALTITUDE

CONTROLLING OBSTACLE (ID/NAME)	
OBSTACLE HEIGHT (ft)	
MOC (ft)	
REQUIRED HEIGHT (ft)	

SHORTEST DISTANCE TO OBSTACLE (ft) : Dr+Do	
PUBLISHED PDG (%) : Max PDG	
DIST. FOR OBST AVOIDANCE (ft) $\frac{\text{Required HGT} - 16\text{ft} - (Dr + Do) * 3.3\%}{\text{Published PDG} - 3.3\%}$	
ALTITUDE FOR OBST AVOIDANCE (ft) Distacne for OBST avoidance * Published PDG + 16ft + DER ELEV	
DETERMINE PDG & ALTITUDE	
ATC PURPOSE	
6. CHECK THE LENGTH OF TIA(TURN INITIATION AREA)	
TYPE OF TURN (Turn at FIX, Turn at Altitude)	
MINIMUM HEIGHT OF TURN POINT (ft)	
PUBLISHED PDG (%)	
REQUIRED DISTANCE FROM DER TO TURN POINT (NM)	
DESIGNED DISTANCE FROM DER TO TURN POINT (NM)	
ESTIMATED HEIGHT AT TURN POINT (ft)	
CONFIRMATION	

8. RNAV SID(DEP)

 국토교통부 Ministry of Land, Infrastructure and Transport		RNAV DEPARTURE PROCEDURE					
1. TITLE OF PROCEDURE		2. RWY NUMBER			3. REVISION DATE		
4. OPERATING AGENCY		5. NEW OR REVISED			6. APPROVAL DATE		
7. AD INFORMATION							
AD NAME	ICAO IDENTIFIER	AD ELEV (ft)	ARP LAT/LONG	ARP MAG VAR			
8. USE RWY INFORMATION							
RWY NO	DIMENSION (m)	LAT/LONG	THR ELEV (ft)	BEARING(T°/M°)			
9. SPECIFICATION & USE SENSOR							
SPECIFICATION	GNSS	DME/DME	VOR/DME				
10. MAP REFERENCE							
PUBLISHING OFFICE	DATE OF PUBLICATION	MAP NAME	SCALE				
11. PROCEDURE & REVISION SUMMARY							
12. PLAN VIEW OPERATIONAL NOTES							
13. DEPARTURE ROUTE DESCRIPTION							
14. FIX DATA							
NAME	TYPE	FROM ARP	ATT	XTT	BV	1/2 WIDTH	LAT/LONG
NOTE							
15. TRACK & SEGMENT DATA							
NAME	PATH DESCRIPTOR	TRK °M(°T)	DIST(NM)	ALT(ft)	SPEED(kt)		
NOTE							
16. MINIMUM STABILIZATION DISTANCE(MSD)							

SEGMENT	DESIGN DIST(NM)	MSD REQUIRED(NM)	REMAIN(NM)	CONFIRMATION

17. TURN PARAMETER

IDENT	TURN ANLGE (")	RADIUS OF TURN[r] (NM)	WIND EFFECT [E90"] (NM)	EARLIEST TURN POINT (NM)	LATEST TURN POINT (NM)

18. CONTROLLING OBSTACLES

ID	NAME	CODE	TYPE	ALTITUDE (ft)	LAT/LONG

NOTE			
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17. MINIMUM SECTOR ALTITUDE (MSA)

NAVAID/WPT			TYPE			IDENT		
TO	FROM	ARC (NM)	ID	NAME	ALT (ft)	LAT/LONG	MOC (ft)	MSA (ft)

NOTE			
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18. EXEMPTION CRITERIA

REQUIRED		REASON	-
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NOTE			
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19. FLIGHT CHECK INFORMATION

DATE OF FLIGHT CHECK	REASON	REMARKS

20. CONCLUSION

21. DEVELOPED BY

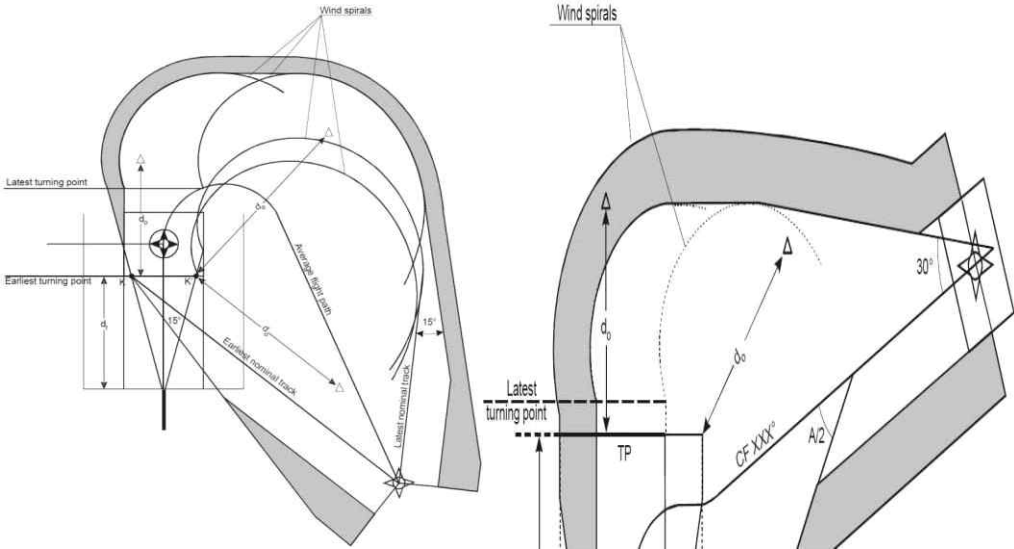
NAME		DATE	

22. APPROVED BY

NAME		DATE	
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23. REMARKS

Serial Number	Path Descriptor	Waypoint Identifier	Fly-over	Course /Track °M(°T)	Distance (NM)	Turn Direction	Altitude (ft)	Speed (kt)	Coordinates	VPA /RDH	Navigation specification	Remarks
1												
2												
3												
4												

APPLICATION OF DEPARTURE PROCEDURE	
NAME OF AIRPORT	INSTRUMENT PROCEDURE
RWY NUMBER & DEPARTURE END OF RUNWAY ELEVATION	
1. EVALUATION RELATIONSHIPS	
	
2. DEPARTURE ROUTE DESCRIPTION	
3. OBSTACLE ANALYSIS BEFORE TURN POINT	
CONTROLLING OBSTACLE (ID/NAME)	
OBSTACLE HEIGHT (ft)	
OIS HEIGHT AT DER (ft) :16ft	
DISTANCE FROM DER TO OBSTACLE (ft)	
MOC (ft) : Distance from DER to OBST * 0.8%	
REQUIRED HEIGHT AT OBST LOCATION (ft) : OBST HGT + MOC	
MINIMUM PDG FOR OBST AVOIDANCE (%)	
[OBST HGT + MOC - OIS Height at DER] / Distance	
4. OBSTACLE ANALYSIS AFTER TURN POINT	
CONTROLLING OBSTACLE (ID/NAME)	
OBSTACLE HEIGHT (ft)	
OIS HEIGHT AT DER (ft) :16ft	

DISTANCE FROM DER TO KK' LINE : Dr (ft)		
SHORTEST DISTANCE FROM KK' LINE TO OBST : Do (ft)		
① (Dr+Do)*0.8% (ft) : MAX 1000ft (Mountainous area 2000ft)		
② MOC FOR TURNING (ft)		
SELECT MOC (ft) : MAX ①, ②		
REQUIRED HEIGHT AT OBST LOCATION (ft) : OBST HGT + MOC		
MINIMUM PDG FOR OBST AVOIDANCE (%) : [OBST HGT + MOC - OIS Height at DER] / Distance		
5. DETERMINE CLIMB GRADIENT AND ALTITUDE		
CONTROLLING OBSTACLE (ID/NAME)	BEFORE TURN POINT	AFTER TURN POINT
OBSTACLE HEIGHT (ft)		
MOC (ft)		
REQUIRED HEIGHT (ft)		
SHORTEST DISTANCE TO OBSTACLE (ft) : Dr+Do		
PUBLISHED PDG (%) : Max PDG		
DIST. FOR OBST AVOIDANCE (ft)	$\frac{\text{Required HGT} - 16\text{ft} - (Dr + Do) * 3.3\%}{\text{Published PDG} - 3.3\%}$	
ALTITUDE FOR OBST AVOIDANCE (ft)		
Distance for OBST avoidance * Published PDG + 16ft + DER ELEV		
DETERMINE PDG & ALTITUDE		
ATC PURPOSE		
6. CHECK THE LENGTH OF ICA		
fly-by waypoint WP1 A1 ATT S DER Minimum 3.5 km (1.9 NM). When the PDG is higher than 3.3 per cent see Part I, Section 3, Chapter 3, 3.3.2.4 flyover waypoint WP1 ATT DER Minimum 3.5 km (1.9 NM). When the PDG is higher than 3.3 per cent see Part I, Section 3, Chapter 3, 3.3.2.4		
TYPE OF TURN (Turn at FIX, Turn at Altitude)		
MINIMUM HEIGHT OF TURN POINT (ft)		
PUBLISHED PDG (%)		
REQUIRED DISTANCE FROM DER TO ETP (NM)		
DESIGNED DISTANCE FROM DER TO ETP (NM)		
ESTIMATED HEIGHT AT TURN POINT (ft)		
CONFIRMATION		

9. CONV STAR

 국토교통부 Ministry of Land, Infrastructure and Transport		CONVENTIONAL ARRIVAL PROCEDURE				
1. TITLE OF PROCEDURE		2. RWY NUMBER			3. REVISION DATE	
4. OPERATING AGENCY		5. NEW OR REVISED			6. APPROVAL DATE	
7. AD INFORMATION						
AD NAME	ICAO IDENTIFIER	AD ELEV (ft)	ARP LAT/LONG	ARP MAG VAR		
8. USE RWY INFORMATION						
RWY NO	DIMENSION (m)	LAT/LONG	THR ELEV (ft)	BEARING(T°/M°)		
9. USE NAVAID INFORMATION						
NAME	LAT/LONG	ELEV (ft)	MAG VAR			
10. MAP REFERENCE						
PUBLISHING OFFICE	DATE OF PUBLICATION	MAP NAME	SCALE			
11. PROCEDURE & REVISION SUMMARY						
12. PLAN VIEW OPERATIONAL NOTES						
13. ARRIVAL ROUTE DESCRIPTION						
14. TRACK & SEGMENT DATA						
NAME	MAG TRK(°)	DISTANCE (NM)	ALT(ft)	DESCENT GRADIENT (%)	SPEED(kt)	LAT/LONG
NOTE						
15. TURN PARAMETER						
IDENT	TURN ANLGE (°)	RADIUS OF TURN[r] (NM)	WIND EFFECT [E90°] (NM)	EARLIEST TURN POINT (NM)	LATEST TURN POINT (NM)	
16. CONTROLLING OBSTACLES						
SEGMENT	ID	NAME	CODE	TYPE	ALTITUDE (ft)	LAT/LONG

NOTE							
17. HOLDING INFORMATION							
IDENT	DIRECTION	ID	NAME	LAT/LONG	ELEV (ft)	MOC (ft)	MIN/MAX ALTITUDE (ft) SPEED (kt)
18. MINIMUM SECTOR ALTITUDE (MSA)							
NAVAID/WPT			TYPE			IDENT	
TO	FROM	ARC (NM)	ID	NAME	ALT (ft)	LAT/LONG	MOC (ft) MSA (ft)
NOTE							
19. EXEMPTION CRITERIA							
REQUIRED		REASON					
NOTE							
20. FLIGHT CHECK INFORMATION							
DATE OF FLIGHT CHECK		REASON				REMARKS	
21. CONCLUSION							
22. DEVELOPED BY							
NAME					DATE		
23. APPROVED BY							
NAME					DATE		
24. REMARKS							

[illegible]

[illegible]

10. RNAV STAR

 국토교통부 Ministry of Land, Infrastructure and Transport			RNAV ARRIVAL PROCEDURE				
1. TITLE OF PROCEDURE		2. RWY NUMBER			3. REVISION DATE		
4. OPERATING AGENCY		5. NEW OR REVISED			6. APPROVAL DATE		
7. AD INFORMATION							
AD NAME	ICAO IDENTIFIER	AD ELEV (ft)	ARP LAT/LONG	ARP MAG VAR			
8. USE RWY INFORMATION							
RWY NO	DIMENSION (m)	LAT/LONG	THR ELEV (ft)	BEARING(T°/M°)			
9. SPECIFICATION & USE SENSOR							
SPECIFICATION	GNSS	DME/DME		VOR/DME			
10. MAP REFERENCE							
PUBLISHING OFFICE	DATE OF PUBLICATION	MAP NAME		SCALE			
11. PROCEDURE & REVISION SUMMARY							
12. PLAN VIEW OPERATIONAL NOTES							
13. ARRIVAL ROUTE DESCRIPTION							
14. FIX DATA							
NAME	TYPE	FROM ARP	ATT	XTT	BV	1/2 WIDTH	LAT/LONG
15. TRACK & SEGMENT DATA							
NAME	PATH DESCRIPTOR	TRK °M(°T)	DIST(NM)	ALT(ft)	DESCENT GRADIENT (%)	SPEED(kt)	
NOTE							
16. MINIMUM STABILIZATION DISTANCE(MSD)							
SEGMENT	DESIGN DIST(NM)	MSD REQUIRED(NM)	REMAIN(NM)	CONFIRMATION			

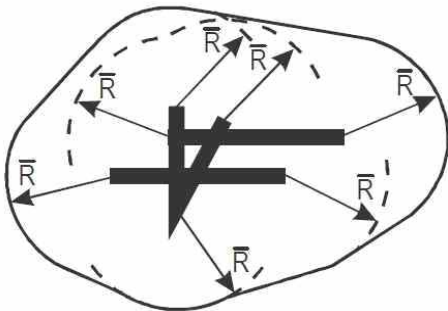
17. TURN PARAMETER									
IDENT	TURN ANLGE(°)		RADIUS OF TURN[r] (NM)		WINDEFFECT E90°](NM)		EARLIEST TURN POINT (NM)		LATEST TURN POINT (NM)
18. CONTROLLING OBSTACLES									
SEGMENT	ID	NAME		CODE	TYPE	ALTITUDE (ft)		LAT/LONG	
NOTE									
19. HOLDING INFORMATION									
IDENT	DIRECTION	ID	NAME		LAT/LONG	ELEV (ft)	MOC (ft)	MIN/MAX ALTITUDE (ft)	SPEED (kt)
20. MINIMUM SECTOR ALTITUDE (MSA)									
NAVAID/WPT				TYPE				IDENT	
TO	FROM	ARC (NM)	ID	NAME		ALT (ft)	LAT/LONG	MOC (ft)	MSA (ft)
NOTE									
21. EXEMPTION CRITERIA									
REQUIRED		REASON	-						
NOTE									
22. FLIGHT CHECK INFORMATION									
DATE OF FLIGHT CHECK		REASON						REMARKS	
23. CONCLUSION									
24. DEVELOPED BY									
NAME						DATE			
25. APPROVED BY									
NAME						DATE			
26. REMARKS									

					AERONAUTICAL DATA TABULATION							
RNAV(GNSS) RWY 07												
Serial Number	Path Descriptor	Waypoint Identifier	Fly-over	Course/Track °M(°T)	Distance (NM)	Turn Direction	Altitude (ft)	Speed (kt)	Coordinates	VPA/RDH	Navigation specification	Remarks
1												
2												
3												
4												
HOLDING PROCEDURE												
Holding Identification	Path Descriptor	Waypoint Identifier	Fly-over	Course/Track °M(°T)	Time (min)	Turn Direction	Altitude (ft)	Speed (kt)	Coordinates	VPA/RDH	Navigation specification	Remarks

			APPLICATION OF ARRIVAL(RNAV) PROCEDURE						
NAME OF AIRPORT		INCHEON INTL			INSTRUMENT PROCEDURE		GUKDO 1A		
1. MINIMUM ALTITUDE COMPUTATION									
SERIAL NUMBER	1	2	3	4	5	6	7	8	
SEGMENT									
SEGMENT LENGTH (NM)									
COURSE/TRACK °M(°T)									
OBS (ID/NAME)									
COORDINATION (LAT/LONG)									
OBS ELEVATION (ft)									
OBS LOCATION (P or S)									

MOC (ft)								
UNCORRECTED MOCA (ft)								
PULISHED MOCA (ft)								
ALTITUDE RESTRICTION (ft)								
SPPED RESTRICTION (kt)								

11. CIRCLING APPROACH COMPUTATION

 국토교통부 Ministry of Land, Infrastructure and Transport		CIRCLING APPROACH COMPUTATION			
NAME OF AIRPORT		INSTRUMENT PROCEDURE			
1. COMPUTATION DATA					
AERODROME ELEVATION(ft)	TEMPERATURE	WIND FACTOR(kt)		BANK ANGLE(°)	
2. RADIUS(R) FOR VISUAL MANOEUVRING (CIRCLING) AREA					
Category A aircraft $\bar{R} = 3.16 \text{ km}$ or 1.68 NM (at 600 m (2 000 ft) MSL) 					
CATEGORY OF AIRCRAFT	A	B	C	D	
CIRCLING SPEED (kt) : IAS					
AERODROME ELEVATION + 1000ft (ft)					
CIRCLING SPEED (kt) : TAS + 25kt Wind factor					
RADIUS (r) OF TURN (NM) : Using Bank angle 20°					
RADIUS (r) OF TURN (NM) : Using Turn rate of 3° per sec					
STRAIGHT SEGMENT (NM)					
RADIUS (R) FROM THRESHOLD (NM)					
3. MDA/H COMPUTATION					
CATEGORY OF AIRCRAFT	A	B	C	D	
CONTROLLING OBSTACLE (ID/NAME)					
OBSTACLE ELEVATION (ft)					
MOC (ft)					
MDA ADJUSTMENTS (ft)					
UNCORRECT ADJUSTED MDA (ft)					
CORRECTED MDA (ft)					
LOWER LIMIT FOR MDA BY PANS-OPS					
MDA FOR STRAIGHT IN APPROACH : LNAV RWY 25					
PUBLISHED MDA (ft)					
4. VISIBILITY COMPUTATION					
CATEGORY OF AIRCRAFT	A	B	C	D	
REQUIRED VIS FOR MAPt TO RWY THR DISTANCE (m)					
VISIBILITY FROM ICAO DOC 8168 TABLE 1-4-7-3 (m)					
VISIBILITY FROM ICAO DOC 9365 TABLE 6-3 (m)					
VISIBILITY FROM ICAO DOC 9365 TABLE 6-1 (m)					
PUBLISHED VISIBILITY (m) : Highest from above					
5. PUBLISHED MINIMA					

CATEGORY		A	B	C	D
CIRCLING	MDA(H)				
	VISIBILITY				
6. REMARKS					

12. CONV ATS ROUTE

 국토교통부 Ministry of Land, Infrastructure and Transport		CONVENTIONAL ATS ROUTE					
1. ROUTE DESIGNATOR		2. TYPE OF ROUTE			3. REVISION DATE		
4. OPERATING AGENCY		5. NEW OR REVISED			6. APPROVAL DATE		
7. NAVIGATIONAL AIDS USED							
NAVAID TYPE	ID	FREQ	CHAN	ANT ELEV (ft)			
8. MAP REFERENCE							
PUBLISHING OFFICE	DATE OF PUBLICATION		MAP NAME		SCALE		
9. PROCEDURE & REVISION SUMMARY							
10. FIX DATA							
REFERENCE DATE OF MAG VAR				2015. 01. 01. (WMM 2015)			
SERIAL NUMBER	FIX NAME		LAT/LONG		MAG VAR (°W)		
NOTE							
11. TRACK & SEGMENT DATA							
SEGMENT	TRK °M-1	TRK °M-2	DIST (NM)	COP	AIRSPACE CLASSIFICATION	MEA (ft)	
NOTE							
12-1. PROTECTION AREA (NORMAL TURN METHOD)							
FIX	PRIMARY(NM)	BUFFER(NM)	TURN ANGLE(°)	RADIUS OF TURN[r](NM)	VALUE OF T (NM)	EARLIEST LIMIT (NM)	LATEST LIMIT (NM)
12-2. PROTECTION AREA (REFINED TURN METHOD)							
FIX	PRIMARY(NM)	BUFFER(NM)	TURN ANGLE(°)	RADIUS OF TURN[r](NM)	WIND EFFECT (E90°)	EARLIEST LIMIT (NM)	LATEST LIMIT (NM)
13. CONTROLLING OBSTACLE DATA							
SEGMENT	OBS NAME	CODE	TYPE	ELEV (ft)		LAT/LONG	

NOTE					
14. EXEMPTION CRITERIA					
REQUIRED		REASON	-		
NOTE					
15. FLIGHT CHECK INFORMATION					
DATE OF FLIGHT CHECK		REASON			REMARKS
16. DEVELOPED BY					
NAME				DATE	
17. APPROVED BY					
NAME				DATE	
18. REMARKS					

		APPLICATION OF ATS(CONV) ROUTE						
OPERATING AGENCY				INSTRUMENT PROCEDURE				
1. MINIMUM ALTITUDE COMPUTATION								
SERIAL NUMBER	1	2	3	4	5	6	7	8
SEGMENT								
CONTROLLING OBS NAME								
OBS ELEVATION (ft)								
TERRAIN CLASSIFICATION N=Normal, M=Mountainous								
OBS LOCATION (P or S)								
MOC (ft)								
UNCORRECTED MOCA (ft)								
PUBLISHED MOCA (ft)								
ALTITUDE RESTRICTION (ft)								

13. RNAV ATS ROUTE

 국토교통부 Ministry of Land, Infrastructure and Transport		RNAV ATS ROUTE				
1. ROUTE DESIGNATOR		2. TYPE OF ROUTE			3. REVISION DATE	
4. OPERATING AGENCY		5. NEW OR REVISED			6. APPROVAL DATE	
7. SPECIFICATION & SENSOR						
SPECIFICATION	GNSS		DME/DME		VOR/DME	
RNAV 2						
RNAV 5						
8. MAP REFERENCE						
PUBLISHING OFFICE	DATE OF PUBLICATION		MAP NAME		SCALE	
9. PROCEDURE & REVISION SUMMARY						
10. FIX DATA						
REFERENCE DATE OF MAG VAR			2015. 01. 01. (WMM 2015)			
SERIAL NUMBER	FIX NAME	VOR/DME BRG & DIST		LAT/LONG	MAG VAR (°W)	
NOTE						
11. TRACK & SEGMENT DATA						
SEGMENT	NAVIGATION SPECIFICATION	TRK °M-1	TRK °M-2	DIST (NM)	AIRSPACE CLASSIFICATION	MEA (ft)
NOTE						
12. MINIMUM STABILIZATION DISTANCE (MSD)						
SEGMENT	DESIGN DIST	MSD REQUIRED	REMAIN		CONFIRMATION	
13. PROTECTION AREA						
WAYPOINT	SEMI WIDTH (NM)	TURN ANGLE(°)	RADIUS OF TURN[r](NM)	WIND EFFECT (E90°)	EARLIEST POINT (NM)	LATEST POINT (NM)
13. CONTROLLING OBSTACLE DATA						
SEGMENT	OBS NAME	CODE	TYPE	ELEV (ft)	LAT/LONG	

NOTE			
14. EXEMPTION CRITERIA			
REQUIRED		REASON	-
NOTE			
15. FLIGHT CHECK INFORMATION			
DATE OF FLIGHT CHECK	REASON		REMARKS
16. DEVELOPED BY			
NAME			DATE
17. APPROVED BY			
NAME			DATE
18. REMARKS			

		APPLICATION OF ATS(RNAV) ROUTE						
OPERATING AGENCY				INSTRUMENT	PROCEDURE			
1. MINIMUM ALTITUDE COMPUTATION								
SERIAL NUMBER	1	2	3	4	5	6	7	8
SEGMENT								
CONTROLLING OBS NAME								
OBS ELEVATION (ft)								
TERRAIN CLASSIFICATION N=Normal, M=Mountainous								
OBS LOCATION (P or S)								
MOC (ft)								
UNCORRECTED MOCA (ft)								
PUBLISHED MOCA (ft)								
PUBLISHED MEA (ft)								